What Is This Course About?

One common challenge senior engineers face is the lack of experience in designing scalable systems. The reason because it is not easy to get an opportunity to work on a large project, especially from the ground up. Most of the time, software engineers get to work on a small part of a bigger system. For these reasons, most engineers feel less prepared for system design interviews, as they lack the adequate knowledge required for designing large systems. This course is created to help developers learn key system design skills that will help them in interviews and in their professional careers.

One way to improve software designing skills is to understand the architecture of famous systems. This is equivalent to learning from others who have worked on designing large systems. In our experience, if a developer has a good understanding of the architecture of a complex system, it becomes pretty easy for them to gain knowledge from other systems even in a different domain. This is true because most system design techniques can easily be adapted and applied to other distributed systems.



What to expect

The course has two parts: System Design Case Studies and System Design Patterns.

Part 1: System design case studies

In the first part, we will go through the architecture of a carefully chosen set of distributed systems:

- Key-value store: Dynamo
- No-SQL wide column stores: Cassandra and BigTable
- Distributed messaging and streaming system: Kafka
- Distributed file storage systems: GFS and HDFS
- Distributed coordination and locking service: Chubby (similar to Zookeeper)

Part 2: System design patterns

In the second part of this course, we will describe a set of design problems (and their solutions) that are common to distributed systems. We call these techniques 'System Design Patterns,' as they can be applied to all kinds of distributed systems and are very handy, especially in a system design interview. A few examples of such patterns are:

1. Write-ahead logging
2. Bloom filters
3. Heartbeat
4. Quorum
5. Checksum
6. Lease
7. Split Brain

Happy learning!
Design Guru's team